

MFMS Week 8 Project Report

Course: PAP521S Programming in Practice **Project:** Municipal Financial Management System (MFMS) **Group:** [Group name or number] **Date:** [Date]

1. Group Members and Contributions

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The MFMS is a program that helps a municipality keep track of its employees, suppliers and money. It runs in the terminal and is controlled by a menu. This week we joined our separate modules into one program and moved repeated code into reusable functions.

3. Modules

Main Menu. Shows the list of options and calls the matching function. If the user types something invalid, the program says so and shows the menu again.

Employee Search. The user enters an employee ID or name and the program shows the matching employee, or a message if nobody matches.

Supplier Management. The user can add a supplier with a name, email, phone number and town, show all saved suppliers, and search for a supplier by name. Up to five suppliers can be stored.

Salary Calculation. The user enters the basic salary, housing allowance and transport allowance, and the program adds them to give the gross salary.

VAT Calculation. The user enters an amount and the program calculates the VAT on it.

Budget Calculation. The user enters revenue and expenses. The program subtracts one from the other and reports a surplus, a deficit or a balanced budget.

4. How the Program Was Built

The program is split into small functions, each with one job, instead of putting everything in `main()`. This makes the code easier to read, test and fix. The Supplier Management module is kept in its own files (`supplier.c` and `supplier.h`) so it could be developed separately and then added to the main program.

5. Testing

The program was compiled with GCC and tested by:

- adding, displaying and searching suppliers

- searching for a supplier that does not exist
- entering invalid menu choices
- checking the salary, VAT and budget results by hand

[Add anything else your group tested.]

6. Challenges and What We Learned

We used functions together with other basic C programming concepts and syntax (character arrays, string functions, loops, `if` statements and menus with `switch`) to build, test and operate this program. Working through these pieces showed us how breaking a program into small functions makes it easier to find and fix mistakes.

7. Conclusion

The MFMS now works as one menu-driven program with its modules connected through reusable functions. In the coming weeks it can be extended further.

Repository: [<https://github.com/225173131-Christian/MFMS-PAP-Project.git>]